



SEAT No. _____

No of Printed Pages: 02

[153]

SARDAR PATEL UNIVERSITY

S.Y.B.C.A. 4th SEM CBCS (Reg & NC) Examination April-2022

US04SBCA25 || Information Security

Date: 08/04/2022 Friday

Time: 03:00pm to 04:00pm

Maximum Marks: 35

Q.1 Answer the following questions.

[05]

1. _____ Attacks are in the nature of eavesdropping on, or monitoring of transmissions.
(a) Passive (b) Active
(c) Threat (d) None of these
2. _____ cipher process the input elements continuously, producing output one element at a time.
(a) Block (b) Stream
(c) Feistel (d) None of above
3. _____ is code embedded in some legal program that is set to "explode" when certain conditions are met.
(a) Worms (b) Back doors
(c) Logic bomb (d) Virus
4. A Packet filter firewall filters at the network or _____ layer.
(a) Physical (b) Application
(c) Data link (d) Transport
5. A _____ configuration involves stand-alone firewall devices plus host based firewalls working together under a central administrative control.
(a) Distributed firewall (b) Packet-filter firewall
(c) Proxy level (d) Circuit-level

Q.2 Fill in the blanks and True/False.

[04]

1. The _____ service is concerned with assuring that a communication is Authentic.
2. _____ is the original message or data that is fed into the algorithm as input.
3. E-mail virus sends itself to everyone on the mailing list in the user's e-mail Package. (True/False)
4. Cryptography is a device installed between the internal network of an organization and the rest of the Internet. (True/False)

[P.T.O.]

Q.3 Explain following in brief. (Attempt any Five)

[10]

- 1 What is Computer Security? Explain in short.
- 2 What is Threat & Attack? Explain in short.
- 3 Explain use of Secret-key.
- 4 Explain Symmetric-key (Secret-key) cryptography in short.
- 5 List out Intruders classes? Explain any one.
- 6 What is Worm? Explain in short.
- 7 Explain four general techniques that firewalls use to control access.
- 8 Explain capabilities (scope) of a firewall.

Q.4 Explain the following in detail. (Attempt any Four)

[16]

- 1 Explain Security Attacks in Detail.
- 2 Explain Security Services in Detail.
- 3 Explain Introduction to Cryptography in detail.
- 4 Explain Substitution & Transposition ciphers in detail.
- 5 Explain Nature of viruses in detail.
- 6 Explain Antivirus approaches in detail.
- 7 Explain Design Principles Firewalls.
- 8 Explain Digital Signature in detail.

* * * * *